



KLH
BACHUPALLY

STUDENT EXAM SCORE PREDICTOR

Python-Based Academic Performance Prediction (Without Machine Learning)

SECOND REVIEW

Name of guide :
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1. Project Title & Team Details

Review 2 • Team and project context

TEAM

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Project Title

Student Exam Score Predictor

Domain

Machine Learning • Educational Data Mining • Learning Analytics

Institution

KLH Bachupally Campus, Hyderabad

2. Problem Statement

Why exam-score prediction is useful in higher education

PROBLEM

Core Problem

Students' final exam outcomes are influenced by multiple academic and behavioral factors. Traditional evaluation often identifies weak performance only after marks are published.

Real-World Relevance

An early prediction can help faculty identify at-risk learners, provide targeted support, and make academic interventions more timely.

Motivation

Use existing student information—such as previous marks, attendance, study time, failures and assessment scores—to estimate likely exam performance.

Research Basis

Recent educational-data-mining studies show that ML models can predict academic outcomes and support early-warning systems, while interpretability and fairness remain important gaps.

3. Objectives

Specific objectives and expected outcomes

OBJECTIVES

01

Collect & prepare data

Build a clean dataset from academic and relevant student attributes.

02

Identify important factors

Study which variables contribute most to exam-score prediction.

03

Train ML models

Compare suitable regression/classification algorithms.

04

Evaluate performance

Use MAE, RMSE, R^2 and/or accuracy , precision, recall and F1 as appropriate.

05

Generate useful predictions

Provide an estimated score or performance category for early academic support

06

Support intervention

Present interpretable insights rather than treating the prediction as a final judgment.

4. Literature Survey / Existing System

Key research findings relevant to the proposed project

LITERATURE

Tomasevic et al. (2020)

Compared supervised data-mining techniques for student exam performance; neural networks showed strong overall results.
Computers & Education, 143, 103676. DOI: 10.1016/j.compedu.2019.103676

Huang et al. (2021)

Systematic review of Student Performance Prediction (SPP): data collection → modeling → prediction → application; highlights personalized education.
Frontiers in Psychology, 12, 698490. DOI: 10.3389/fpsyg.2021.698490

Smart Learning Environments (2022)

Compared Random Forest, NN, SVM, Logistic Regression, Naïve Bayes and kNN for final-exam prediction; midterm grades were important predictors.
Smart Learning Environments. DOI: 10.1186/s40561-022-00192-z

Khairy et al. (2024)

Compared Random Forest, Decision Tree, Naïve Bayes, Neural Network and KNN; RF and DT performed strongly for student performance classification.
Education and Information Technologies, 29, 21621–21645. DOI: 10.1007/s10639-024-12619-w

Existing System & Limitations

What current approaches do—and where the gap remains

GAP

Existing Approaches

- Manual analysis of marks and attendance
- Rule-based identification of weak students
- Python programs using structured student records
- Weighted-score calculation for estimated final marks and performance bands
- Rule-based thresholds for early identification of students needing support

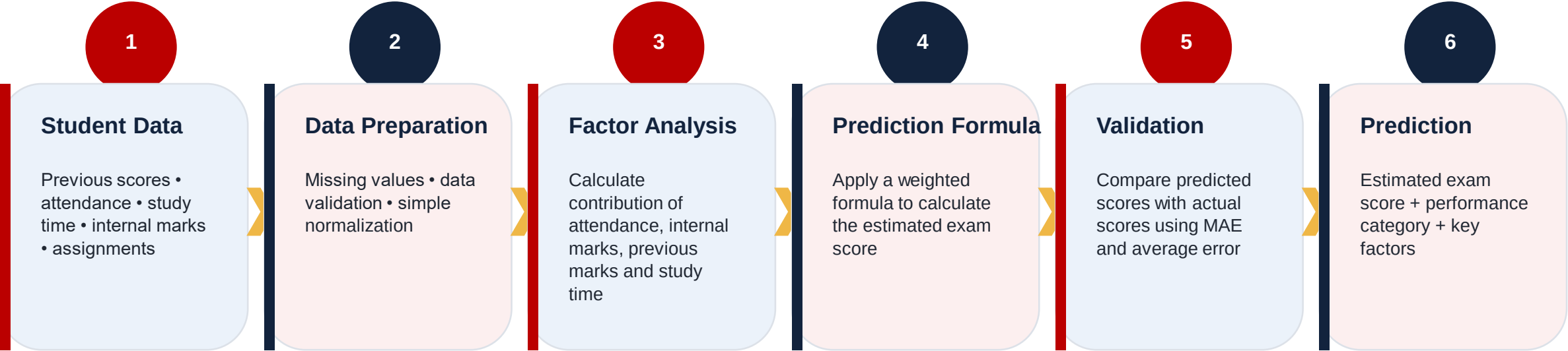
Limitations / Research Gaps

- Manual systems can be slow and difficult to maintain for many students
- Results may not generalize to a specific university or course
- A simple formula is transparent, so students and faculty can see how the result was obtained
- Fairness and bias can be overlooked
- Predictions may be used too late, after major assessments

Proposed System

Student Exam Score Predictor – Python rule-based workflow

PROPOSAL



Methodology & Model Plan

A reproducible pipeline for training and evaluating the predictor

METHOD

Data acquisition

Collect academic records and/or use a benchmark dataset such as UCI Student Performance for baseline experiments.

Exploratory analysis

Inspect score distributions and relationships between attendance, study habits, previous grades and outcomes.

Formula validation

Test the formula on past student records and compare estimated scores with actual final scores.

Data cleaning

Handle missing values, duplicates, inconsistent entries and outliers; encode categorical variables.

Create prediction rules

Define weights and score ranges; keep the calculation consistent for every student.

Python implementation

Use pandas for data handling, NumPy for calculations and Streamlit/Tkinter for the user interface.

Dataset & Key Features

Suggested variables for the Student Exam Score Predictor

DATA

Academic

Previous exam marks, internal assessment, assignment scores, previous GPA/SGPA, failures

Engagement

Attendance percentage, LMS activity, assignment completion, class participation

Study habits

Study time, revision frequency, preparation time, learning resources

Assessment

Midterm / internal marks, practical scores, quiz performance

Context

Course, semester, subject, credit load, class section

Output

Final exam score (regression) or pass/fail / grade band (classification)

Expected Outcomes

What the project should deliver by the final review

OUTCOMES

Prediction

Estimate a student's likely exam score or performance band before the final exam.

Early warning

Flag students who may need additional academic support.

Dashboard / UI

Provide a simple interface for entering student inputs and viewing the result.

Prediction validation

Compare predicted scores with actual scores using average error and percentage accuracy.

Easy explanation

Display the input values, weighted contributions and final estimated score.

Actionable support

Use predictions to guide mentoring—not to label or penalize students.

Research Papers to Cite in the Project

Recommended references for the literature survey and report

REFERENCES

- 1. Tomasevic, N., Gvozdenovic, N., & Vranes, S. (2020). “An overview and comparison of supervised data mining techniques for student exam performance prediction.” *Computers & Education*, 143, 103676. DOI: 10.1016/j.compedu.2019.103676.
- 2. Huang, S., Fang, N., & others. (2021). “Educational Data Mining Techniques for Student Performance Prediction: Method Review and Comparison Analysis.” *Frontiers in Psychology*, 12, 698490. DOI: 10.3389/fpsyg.2021.698490.
- 3. “Educational data mining: prediction of students’ academic performance using machine learning algorithms.” *Smart Learning Environments* (2022). DOI: 10.1186/s40561-022-00192-z.
- 4. Khairy, D., Alharbi, N., Amasha, M. A., Areed, M. F., Alkhalaf, S., et al. (2024). “Prediction of student exam performance using data mining classification algorithms.” *Education and Information Technologies*, 29, 21621–21645. DOI: 10.1007/s10639-024-12619-w.
- 5. “Predicting student performance: A comprehensive review of machine learning, deep learning, and explainable AI approaches.” *Computers & Education: Artificial Intelligence*, 10 (2026), 100548. DOI: 10.1016/j.caeai.2026.100548.



Conclusion & Next Steps

Key Takeaway

Student Exam Score Predictor applies educational data mining and machine learning to estimate performance early enough for meaningful academic support.

Next Steps

1. Finalize dataset
2. Clean and explore data
3. Train baseline models
4. Compare metrics
5. Build prediction UI
6. Validate, explain and document results

Thank You

Questions & Discussion

Python Implementation — No Machine Learning

1. Store student data: Use CSV/Excel files with attendance, previous marks, internal marks, assignments and study time.
2. Clean the data: Use pandas to remove duplicates, handle missing values and check valid ranges.
3. Calculate a weighted score: Example: 30% previous marks + 25% internal marks + 20% attendance + 15% assignments + 10% study time.
4. Generate prediction: Convert the weighted score into an estimated final exam score.
5. Classify performance: 80–100: Excellent, 60–79: Good, 40–59: Average, below 40: Needs Support.
6. Show result: Python displays the estimated score, performance category and the factors that affected the result.

$$\text{Predicted Score} = 0.30 \times \text{Previous} + 0.25 \times \text{Internal} + 0.20 \times \text{Attendance} + 0.15 \times \text{Assignment} + 0.10 \times \text{Study}$$

